

PYTHON LAB EXERCISE-4

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Creation of an array:

```
#creation of array  
arr=[1,2,3,4]  
print(arr)
```

```
[1, 2, 3, 4]
```

```
#accessing elements  
print(arr[2])  
print(arr[0])
```

```
3  
1
```

```
#manipulation  
arr.append(5)  
print (arr)  
arr.insert(6,7)  
print(arr)
```

```
[1, 2, 3, 4, 5]  
[1, 2, 3, 4, 5, 7]
```

```
#iteration
for i in arr:
    print(i)
    print("...")
    for i in range(len(arr)):
        print(f"index:{i},value:{arr[i]}")
```

```
1
...
index:0,value:1
index:1,value:2
index:2,value:3
index:3,value:4
index:4,value:5
index:5,value:7
```

```
2
...
index:0,value:1
index:1,value:2
index:2,value:3
index:3,value:4
index:4,value:5
index:5,value:7
```

```
3
...
index:0,value:1
index:1,value:2
index:2,value:3
index:3,value:4
index:4,value:5
index:5,value:7
```

```
#2d array
mat=[
    [1,2,3],
    [4,5,6],
    [7,8,9]
]
for i in range(len(mat)):
    for j in range(len(mat[i])):
        print(i , j , mat[i][j])
```

```
0 0 1
0 1 2
0 2 3
1 0 4
1 1 5
1 2 6
2 0 7
2 1 8
2 2 9
```

```
#aarray module
import array
arr=array.array('i',[1,2,3,4,5])
print(arr)

array('i', [1, 2, 3, 4, 5])
```

```
#dosent support all list mehtods
arr.append(50)
print(arr)
arr=array.array('i',sorted(arr))
print(arr)
newarr=arr.tolist()
print(newarr)

array('i', [1, 2, 3, 4, 5, 50])
array('i', [1, 2, 3, 4, 5, 50])
[1, 2, 3, 4, 5, 50]
```

Numpy:

Basic mumpy array:

```
#basic numpy array
import numpy as np
arr1=np.array([1,2,3,4])
print(arr1)
arr2d=np.array([
    [1,2],[3,4]])
print(arr2d)
```

```
[1 2 3 4]
[[1 2]
 [3 4]]
```

Basic numpy manipulation:

```
#Basic array manipulation
import numpy as np
arr1=np.array([1,2,3,4])
print(arr1.shape)
print(arr2d.shape)
print("****")
print(arr1)
print(arr1.reshape(2,2))
print("****")
print(arr1.shape)
print(arr1.ndim)
print(arr1.size)
print(arr1.dtype)
print("****")
print(arr2d.shape)
print(arr2d.ndim)
print(arr2d.size)
print(arr2d.dtype)
```

```
(4,)
(2, 2)
****
[1 2 3 4]
[[1 2]
 [3 4]]
****
(4,)
1
4
int64
***
(2, 2)
2
4
int64
```

Accessing elements:

```
#accessing elemnets
import numpy as np
a2=np.array([
    [1,2,7],[3,4,8],[5,6,9]])
print("accessing")
print(a2[2][1])
print(a2[0:2])
print("accessing columns and rows")
print(a2[:,2])

accessing
6
[[1 2 7]
 [3 4 8]]
accessing columns and rows
[7 8 9]
```

Sum of row and column:

```
#sum
import numpy as np
#modify
#sum of all elements
sum=np.sum(a2)
print(sum)
#row sum
row_sum= np.sum(a2, axis=1)
print(row_sum)
#column sum
sum= np.sum(a2, axis=1)
print(sum)

45
[10 15 20]
[10 15 20]
```

Built in functions:

```
#built in
a=[1,5,6]
print(np.sum(a))
print(np.mean(a))
print(np.median(a))
print(np.max(a))
print(np.min(a))
```

```
12
4.0
5.0
6
1
```

Mathematical operations:

```
#mathematical operations
a=np.array([1,2,3])
print(a+2)
print(a*3)
```

```
[3 4 5]
[3 6 9]
```

```

a=np.zeros((2,3))
b=np.ones((2,3))
c=np.eye(3)
d=np.arange(0,10,2)
e=np.linspace(0,1,5)
print(a)
print(b)
print(c)
print(d)
print(e)

```

```

[[0. 0. 0.]
 [0. 0. 0.]]
[[1. 1. 1.]
 [1. 1. 1.]]
[[1. 0. 0.]
 [0. 1. 0.]
 [0. 0. 1.]]
[0 2 4 6 8]
[0.  0.25 0.5  0.75 1.  ]

```

Flattening:

```

#flattening
import numpy as np
a=np.array([1,2,3,4,5,6])
b=a.reshape(2,3)
c=b.flatten()
d=np.append(c,9)
e=np.delete(c,3)
print(d,e)

```

```

[1 2 3 4 5 6 9] [1 2 3 5 6]

```


Addition of 2 arrays:

```
#addition of 2 arrays
import numpy as np

# Create NumPy arrays
arr1 = np.array([1, 2, 3, 4])
arr2 = np.array([5, 6, 7, 8])

# Add them directly
result = arr1 + arr2

print(f"NumPy Result: {result}")

NumPy Result: [ 6  8 10 12]
```

Addition of 2 matrices:

```
import numpy as np

# Create NumPy arrays
A = np.array([[1, 2, 3], [4, 5, 6], [7, 8, 9]])
B = np.array([[9, 8, 7], [6, 5, 4], [3, 2, 1]])

# Add matrices directly
result = A + B

print(result)

[[10 10 10]
 [10 10 10]
 [10 10 10]]
```